QUEST: User-Based Quality of Service Aware Uplink Resource Scheduling

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***Abstract*—Efficient radio resource management (RRM) in 5G networks is increasingly challenged by the diverse quality of service (QoS) requirements of emerging applications and the growing uplink (UL) traffic from resource-constrained devices. Existing scheduling approaches often lack user and service-specific context, limiting their ability to guarantee timely and energy-efficient data transmission, particularly critical for the internet of things (IoT) and mission-critical services. In this work, we introduce *QUEST*, a QoS-aware UL scheduling framework that exploits the 5G QoS model alongside network and device context to efficiently allocate radio resources. Designed and evaluated in an indoor factory environment, *QUEST* supports users with various heterogeneous 5QI services under dynamic multi-user conditions. Evaluation results, validated through both real-world measurements and 3GPP-compliant simulations, show that *QUEST* consistently outperforms traditional channel- and QoS-aware schedulers. It improves QoS compliance, reduces packet drops and serving time, and enhances energy efficiency. For users with stringent QoS demands, measurements show a 13% increase in successfully transmitted packets and a 6.2% reduction in delay for 50% of transmissions, compared to the best-performing baseline. Benchmarking against an optimal scheduler shows that *QUEST* achieves the closest performance among baselines, while maintaining low complexity, making it a practical and scalable solution for 5G and beyond UL RRM.**

***Index Terms*—Radio resource management, quality of service, user context, user satisfaction, energy efficiency, IoTs.**

I. Introduction

THE evolution of communication networks has intensified the radio resource management (RRM) challenge, due to the increasing heterogeneity of connected devices supporting applications with diverse requirements. In particular, the proliferation of connected internet of things (IoT) devices, ranging from resource-constrained sensors in smart cities to sophisticated devices in applications such as smart cameras, and healthcare monitoring [1], has led to an increase in uplink (UL) transmissions. This stems from the user-centric nature of IoTs, which, being solely powered by batteries, must meet stringent energy-efficiency requirements [2]. As a result, they frequently transmit data to (i) offload computationally intensive tasks to mobile edge computing (MEC) servers, saving processing power and performing faster computing [3], [4], [5]; (ii) synchronize with digital twins, ensuring that decisions based on twin data reflect the state of physical devices [4], [6], [7], [8]. Fulfilling stringent quality of service (QoS) requirements of diverse and resource-intensive applications is particularly challenging as UL traffic grows in 5G networks while radio resources remain limited [9].

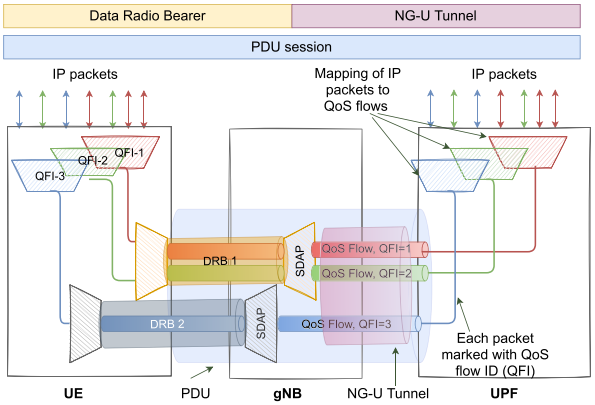


Fig. 1. 5G QoS flow management.

In this context, network slicing [10] provides isolation and static resource allocation for distinct service categories, such as enhanced mobile broadband (eMBB), ultra-reliable and low-latency communication (URLLC), massive machine-type communication (mMTC) [11]. However, slicing alone cannot guarantee the fine-grained user-level QoS [12], especially in heterogeneous networks where diverse traffic profiles coexist within a single slice [13]. Consequently, adaptive, context-aware RRM is essential to dynamically prioritize users and efficiently allocate scarce radio resources, particularly in scenarios where degradation of quality of experience (QoE) or failure to meet delay thresholds can result in critical consequences, as in autonomous systems.

To support applications' requirements, 5G standardization [13], [14] defines QoS model architecture (see Fig. 1), in which end-to-end flows are established to transport application packets between user equipment (UE) and core network (CN). Although application packets within the same flow receive the same forwarding behavior, the actual forwarding and flow priorities are determined by the medium access control (MAC) schedulers. The latter must be aware of applications' QoS requirements to ensure their fulfillment and user satisfaction.

Received 20 June 2025; revised 29 January 2026 and 9 April 2026; accepted 9 April 2026. Date of publication 20 April 2026; date of current version 28 April 2026. This work was supported by the Bavarian Ministry of Economic Affairs, Regional Development and Energy as part of the project "6G Future Lab Bavaria", and by the Federal Ministry of Research, Technology and Space of Germany (BMFTR) in the programme of "Souverän. Digital. Vernetzt." joint project 6G-life, project identification number 16KIS2414. The associate editor coordinating the review of this article and approving it for publication was J. Wang. *(Corresponding author: Alba Jano.)*

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Digital Object Identifier 10.1109/TNSM.2026.3685537

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Despite the growing interest and research potential on user-level QoS resource scheduling, we argue that current approaches still exhibit critical shortcomings. First, most studies focus on downlink (DL) resource allocation, where context information of environment and network entities is easily accessible [15], [16], [17], [18], [19], [20], [21], [22]. UL scenarios, which are equally or more susceptible to congestion and delay, remain underexplored despite their growing importance for energy-efficient or latency-sensitive applications. Existing UL focused works, include individual metrics such as service priority levels, packet delay budget, number of hybrid automatic repeat request (HARQ) re-transmissions, packet size, or a combination of these metrics. In contrast, *QUEST* jointly accounts for all factors influencing UL scheduling decisions, including traffic urgency, QoS requirements, channel conditions, and buffer dynamics. Second, evaluations of proposed solutions predominantly rely on simulations [20], [21], [22], [23], [24], [25] or have limited validation in real or experimental settings [26]. While simulations abstract several protocol procedures and assume the availability of information in every network component, they often omit the explicit modeling of signaling mechanisms and constraints required to obtain such information in practice. This gap raises concerns about the practical feasibility and robustness of the proposed frameworks under a realistic network environment. *QUEST* addresses this gap by complementing simulations with experimental validations and incorporating signaling solutions to enable its functionality.

Several questions, both from research and practical perspectives, arise related to the efficient UL RRM under the 5G QoS model constraints: (1) Which context information can further enhance the efficiency of UL RRM? (2) How can UEs with diverse QoS requirements be effectively scheduled to maximize their satisfaction? (3) How can device-specific information be integrated into RRM to improve scheduling accuracy? In this work, we address these questions by proposing *QUEST*, an UL QoS-aware resource scheduling framework that leverages not only the environment context but also UEs' context, specifically head-of-line (HOL) delay and operating applications, enabling precise resource allocation tailored to the specific requirements of each UE.

We implement and validate *QUEST* in a 5G software defined network (SDR)-based experimental setup, deployed in an indoor environment with robotic arms and a mobile robot, emulating a small-scale factory scenario based on a private 5G campus network deployment. Our measurements include one receiver (gNB) and multiple transmitters (UEs), each supporting a different application. We compare *QUEST* against representative throughput-aware and QoS-aware schedulers, including priority based (PB) [25], earliest delay first (EDF) [15], and modified-largest weighted delay first (M-LWDF) [18]. Results show that *QUEST* significantly improves QoS compliance while reducing packet loss caused by delay violations or channel conditions. It also reduces the serving time, thereby reducing device wake-up time duration and improving energy efficiency. In measurements involving two applications, *QUEST* achieves, for the higher-priority, more stringent-delay application, a 13% increase in successfully transmitted packets and a 6.2% reduction in delay for 50% of transmissions, compared to the best-performing baseline. Meanwhile, it maintains the QoS compliance for the remaining applications. Simulations with 50 UEs in a 3GPP-compliant simulator [27] confirm the framework's scalability, with simulation results aligning with the measurement data. Furthermore, benchmarking against an optimal Gurobi-based approach demonstrates that *QUEST* achieves the closest performance among all baseline methods, supporting its effectiveness in QoS-aware resource scheduling. Our main contributions in this work are:

- *Enabling new information.* We propose a signaling solution to report UEs' HOL delay to the scheduler, enhancing UEs' context role on the scheduling decisions. The proposed approach leverages the existing buffer status report (BSR) signaling to transmit HOL delay, thereby avoiding the exchange of new signaling messages.
- *New scheduling principle using context information.* We propose *QUEST*, a heuristic QoS-aware resource scheduling approach that leverages the 5G QoS model and prioritizes users based on both environmental context (i.e., channel state information) and enhanced device context, including UE HOL delay and the QoS characteristics of associated 5G applications.
- *Optimal QoS-aware resource allocation.* We formulate the QoS-aware scheduling problem as a mixed-integer programming problem and obtain its optimal solution using Gurobi. This optimal solution provides a theoretical upper bound for evaluating *QUEST*'s performance.
- *QoS-aware user satisfaction metric.* We propose a novel user satisfaction metric that quantifies the degree of QoS fulfillment by jointly capturing throughput, delay, and packet success ratio, explicitly incorporating 5G QoS identifier (5QI) parameters to reflect heterogeneous QoS requirements in 5G systems.
- We release the implementation code of the 5G scheduling framework developed in OpenAirInterface (OAI), along with supporting datasets that include environmental information and RRM data for various schedulers, at: https://github.com/tum-lkn/QUEST

The remainder of this work is organized as follows. Section II provides a comprehensive review of related works focused on resource management. We provide relevant background to the topic in Section III. Section IV discusses in detail our system model and optimization problem. This is followed by a detailed description of the measurement setup in Section V. Next, Section VII describes the achieved measurement and simulation results. We discuss possible deployment aspects in Section VIII, while Section IX concludes the paper.

II. Related Work

The introduction of the 5G QoS model has led to extensive research into QoS-aware scheduling and radio resource allocation. Given the importance of selecting appropriate approaches for real-time operation in practical deployments, we categorize existing approaches into three main groups: (1) heuristic methods, (2) reinforcement learning (RL) techniques, and (3) optimization approaches. The survey in [28] provides an

overview of 5G scheduling algorithms, with the focus on metric-based and machine learning (ML) approaches.

Heuristic algorithms such as round robin (RR), maximum throughput (MT) [29], and proportional fair (PF) [30], pursue various objectives, including resource fairness, system throughput maximization, and throughput fairness among users. An implementation in the DL of weighted versions of PF and MT is presented in [26] using an experimental setup. However, these algorithms primarily focus on throughput, overlooking critical performance indicators such as latency, reliability, and energy efficiency. This makes them less suitable for the diverse application requirements of 5G networks. To address latency requirements, the EDF scheduler [15] prioritizes packets approaching their delay thresholds. Subsequent works [16], [17] extend this idea by incorporating payload size into the scheduling metric, targeting time-critical physical systems. They evaluate these approaches through simulations [16] and SDR-based experimental setup [17]. While these algorithms enhance QoS compliance for latency-sensitive traffic, they disregard channel conditions, limiting their adaptability in a dynamic wireless environment.

The M-LWDF algorithm [10], [18] extends PF by incorporating HOL delay, application delay threshold, and the probability of deadline violation. Similarly, focusing on eMBB traffic, [19] proposes a channel- and delay-aware scheduler that prioritizes small packets approaching their delay threshold for users with favorable channel conditions. Although these approaches improve QoS satisfaction, their performance degrades under congestion, particularly when multiple flows share similar delay thresholds but differ in priority. The scheduler in [25] addresses this by incorporating 5G QoS model parameters, including priority levels, packet delay budget, reported channel quality indicator (CQI) value, number of HARQ retransmissions, packet size, and combinations of these metrics. Their findings indicate that relying solely on standardized parameters, such as packet priority, is insufficient to meet the diverse requirements of emerging applications.

To introduce adaptability, their work in [24] proposes a high-level RL framework that dynamically selects among a set of pre-developed heuristic algorithms. Similarly, [22] presents an RL framework, choosing the best-performing scheduler among a group of well-established schedulers. While these frameworks improve flexibility, their performance is bounded by the limitations of the underlying heuristics. To overcome this, several studies [20], [21], [23] explore low-level RL approaches or deep learning techniques [31] that directly learn scheduling policies. Although these methods move towards autonomous network operation with improved performance, they introduce high training complexity, scalability challenges in large and dynamic environments [33], and require continuous retraining. In user-centric IoT scenarios with battery-powered devices and stringent energy and latency constraints, such as dynamic indoor factory environments, deploying DRL-based schedulers is impractical and increases the cost of online training. Consequently, we do not include DRL-based approaches in our comparison and instead benchmark *QUEST* against classical channel and QoS-aware heuristics, enabling evaluation under realistic and deployable conditions.

Optimization-based techniques are also widely employed to achieve near-optimal or sub-optimal resource allocation. For instance, the work in [32] uses Lyapunov optimization in DL to minimize the long-term time average of packet drops subject to queue stability and average service rate guarantee constraints. In [33], Lagrangian optimization is applied to minimize scheduled capacity under packet and QoS level constraints. While these methods are efficient in simulations, their high computational complexity raises concerns for real-time performance in hardware systems.

Despite the advancements, most works primarily focus on DL scheduling and rely on simulations [15], [16], [17], [18], [19], [20], [21], [22], [23], [32], [33]. Although simulations enable the design of sophisticated algorithms without real-time constraints, they often abstract key features such as detailed channel models and protocol stack (e.g., CQI reporting, resource allocation granularity). With the emergence of diverse applications, in this work, we shift focus towards UL QoS-aware scheduling, accounting for the main factors influencing scheduling decisions. We validate the scheduling strategies in an experimental setup while considering the architecture and protocol signaling.

III. Background

This section provides a brief overview of 5G QoS model, resource allocation in the orthogonal frequency division multiple access (OFDMA) system, and BSR signaling to facilitate the understanding of the remainder of the paper.

A. 5G QoS Model

The 5G QoS model [13], [14], [34] is based on the concept of QoS flows, which represent the finest granularity of QoS differentiation in 5G networks. All packets mapped to the same QoS flow, as shown in Fig. 1, receive identical processing and forwarding treatment (e.g., scheduling policy). To describe the packet forwarding treatment between the UE and user plane function (UPF), 5G defines a fixed set of QoS profiles. Each profile is uniquely identified by a 5QI value, ranging from 1 to 90, depending on the service type.

Table 5.7.4-1 in [13] outlines the standardized QoS characteristics associated with each 5QI, including parameters such as resource type (guaranteed bit rate (GBR), non-GBR, delay-critical GBR), maximum data burst volume (MDBV), packet priority level, packet delay budget (PDB), and packet error rate (PER). The packet priority level determines the precedence of flows during resource allocation, enabling differentiation among multiple QoS flows either from the same UE or different UEs. In a congested scenario, higher-priority flows are served first. The PDB defines the maximum tolerable one-way delay from the arrival of a packet at the UE to its delivery at the UPF. The PER specifies the maximum fraction of IP packets that may be lost due to channel conditions (i.e., processed by the UE but not successfully delivered to the gNB). The MDBV specifies the largest amount of data that can be transmitted during PDB.

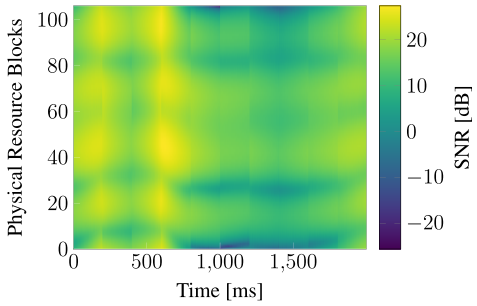


Fig. 2. Illustration of time-frequency selectivity of an indoor factory wireless channel, for a user connected to a gNB operating with 40 MHz system bandwidth in 30 kHz subcarrier spacing, generated in NextGSIM [27].

Upon UE attachment, one or multiple packet data unit (PDU) sessions are established, as shown in Fig. 1. Each session consists of an N3-GTP-U Tunnel (between gNB and UPF) and data radio bearers (DRBs) (between UE and gNB). On the N3 interface, IP packets are mapped to QoS flows based on their QoS requirements. Each packet is marked with a QoS flow identifier (QFI), which ranges from 0 to 63. This identifier may correspond to the 5QI value or be dynamically assigned according to pre-defined mapping rules. Because radio access network (RAN) manages traffic via DRBs, a mapping between QoS flows and DRBs is required. This mapping is not necessarily a one-to-one, as multiple QoS flows can be mapped to the same DRB [14]. To support this mechanism, 5G introduces service data adaption protocol (SDAP) layer, which maps the QoS flows to DRBs according to the QFI markings. The SDAP layer does not include any scheduling capabilities related to the QoS flows.

For the UL traffic, the UE classifies and maps packets based on: (i) explicitly signaled QoS rules, (ii) pre-configured rules in the UE, or (iii) reflective QoS. Explicit QoS rules are configured by the session management function (SMF) during PDU session establishment and define how UL traffic should be handled. In the case of reflective QoS, the UE infers the mapping by observing how DL packets are associated with QoS flows and mirrors that behavior in the UL. In our setup, we assume the use of explicit QoS rule-based mapping.

B. Resource Allocation

The MAC layer is the only layer within the 5G access network that directly supports QoS traffic engineering techniques. At the base station (gNB), radio resource schedulers exploit the time-frequency selectivity of wireless channels, as illustrated in Fig 2, to efficiently allocate available physical resource blocks (PRBs). A PRB represents the smallest allocatable unit of frequency resources at a given time, consisting of 12 consecutive subcarriers over a slot duration. The slot duration scales with the subcarrier spacing, enabling low-latency transmissions as specified in [35]. Efficient scheduling requires timely knowledge of the UE's channel conditions, buffer status, and data flow priorities.

Accurate UL channel knowledge improves both scheduling and link adaptation at the gNB, but it introduces signaling overhead, particularly due to the need for frequent transmission of sounding reference signals (SRS) from the UE [10].

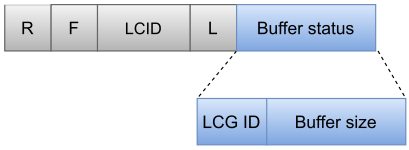


Fig. 3. MAC control element for buffer status reports [39].

In practical deployments, channel state information (CSI) is obtained in either wideband or subband granularity [36]. Wideband CQI provides a single value representing the average channel quality across the entire bandwidth. In contrast, subband CQI captures channel quality over narrower frequency segments, enabling finer frequency-domain granularity. A subband consists of a contiguous set of PRBs, with its size determined by the total number of PRBs in the configured bandwidth part. The UE is configured by higher layers with one of two subband size options as defined in [36]. Subband CQI allows the scheduler to prioritize transmissions in frequency segments with high estimated channel quality, thus improving overall spectral efficiency.

To further reduce signaling overhead, resource allocation is performed in units of resource block groups (RBGs), each typically consisting of 2 to 16 PRBs. The RBG size is configured according to the total number of PRBs within the bandwidth part [36]. Resource allocation is communicated to UEs using either Type 0 or Type 1 signaling. Type 0 uses a bitmap to flexibly indicate assigned RBGs, while Type 1 specifies a contiguous set of PRBs using a start index and length. In this work, we adopt Type 0 allocation to align with subband CQI granularity and better exploit frequency selectivity in the wireless channel.

Based on estimated CQI, the gNB selects a modulation coding scheme (MCS), used for data transmission over the assigned resources. Since the throughput in an orthogonal frequency division multiple (OFDM) system depends on the modulation order, coding scheme, coding rate, and the resulting bit error rate (BER), the gNB employs adaptive modulation and coding (AMC) to optimize UEs' spectral effciency [37]. Regardless of the CSI type, the UE transmits using a single MCS during each scheduling interval [38].

C. Buffer Status Report

The BSR mechanism supports QoS-aware UL scheduling by informing the gNB of the amount of data buffered at the UE. This information is conveyed via a MAC control element (CE), as specified in [39]. To manage signaling overhead efficiently, logical channels (LCs) are grouped into logical channel groups (LCGs), with each UE supporting up to eight LCGs. A BSR can report the buffer status for either a single LCG (Short BSR) or multiple LCGs simultaneously (Long BSR). The Logical Channel ID (LCID) field in the MAC subheader identifies the type of control element being transmitted, including the specific BSR format [39].

Each BSR includes the LCG ID and a buffer size field that indicates the volume of data (in bytes) awaiting transmission for that LCG, as illustrated in Fig. 3. The size of the buffer size field varies depending on the BSR format, typically between

5 and 8 bits, allowing for compact signaling. BSR transmission can be triggered under several conditions, including: (i) the arrival of higher-priority data relative to the currently transmitted LCG, (ii) the expiration of a periodic reporting timer, or (iii) padding optimization, where BSR information is included in unused bits of transport block size (TBS) padding to maximize payload utilization.

In addition, the LCID has a dedicated role in differentiating the signaling and data channels. LCID field values from 0 to 3 are reserved for signaling radio bearers (SRBs), which carry control information and are responsible for transmitting system information, connection management, and QoS configuration. LCIDs values larger than 3 are associated with DRBs, which are the primary focus of this study.

IV. System Model and Problem Formulation

We study an UL OFDM-based 5G network deployed in an indoor factory environment, where a single gNB denoted by B, serves a set of UEs indexed by $i = \{1, 2, \ldots, I\}$. The gNB operates with a bandwidth W and subcarrier spacing σ, determining the total number of available PRBs, denoted by J, where $j = \{1, 2, \ldots, J\}$. According to [36], the PRBs are organized into R RBGs, indexed by $r = \{1, 2, \ldots, R\}$, which are allocated to UEs. At each transmission time interval (TTI), indexed by $t = \{1, 2, \ldots, T\}$, UEs with data to transmit compete for the available UL resources to achieve successful and timely transmission.

We assume that each UE runs a single application characterized by a 5QI, identified by QFI_i, with associated priority level P_i, latency threshold PDB_i, and target packet error rate PER_i (see Section III-A). The traffic generated by each UE i consists of a set of packets K_i, where each packet $k \in K_i$ has a size $pkt_i^{k,bits}$ and an arrival time $pkt_i^{k,t}$. These packets are buffered at the RLC layer before transmission. The waiting time of packet k in RLC buffer is denoted $D_{i,w}^k$, and the total service time is D_i^k. The maximum waiting time experienced by the queued packets at time t corresponds to the waiting time of the first packet in the queue, denoted by $D_{i,HOL}^t$. Due to the short communication range in the factory scenario, transmission delay is considered negligible.

Let $s = \{1, 2, \ldots, S\}$ represent the set of subbands in the system. The experimental setup supports only wideband CQI estimation. Meanwhile, our simulation framework evaluates three granularity levels: wideband CQI ($S = 1$), subband CQI ($S > 1$), and per-PRB CQI ($S = J$). The estimated CQI for UE i in subband s at time t is denoted by $CQI_{i,s}^t \in [1, 15]$. To estimate the amount of data transmitted at RBG r, which is part of the subband s, the $CQI_{i,s}^t$ values are mapped to $MCS_{i,s}^t$ according to the spectral efficiency tables from [36]. Each MCS m, mapped for UE i, at subband s during time t, is characterized by a modulation order $Q_{m,i,s}^t$ and a code rate $CR_{i,s}^t$. The corresponding TBS is computed using either table mapping or a formula-based approach [36], based on the calculated N_{info} value

$$N_{info} = N_{RE} \cdot CR_{i,s}^t \cdot Q_{m,i,s}^t \cdot v, \tag{1}$$

where N_{RE} is the number of available resource elements and v number of MIMO layers. The binary variable $\delta_{i,r}^t \in \{0, 1\}$ indicates whether RBG r is allocated to UE i at time t ($\delta_{i,r}^t = 1$). The total TBS allocated to UE i at time t is given by

$$TBS_i^t = \sum_{r \in R} \delta_{i,r}^t \cdot TBS_{i,r}^t. \tag{2}$$

Although subband-level CQI values are available, each UE can use a single MCS across all allocated resources in a given TTI. Therefore, the exponential effective SNR mapping (EESM) technique [37], [40] is applied to derive an effective SNR that accounts for the channel quality variations across subbands, for UE i

$$SNR_i^{eff} = -\alpha_m \cdot \log\left(\frac{1}{N}\sum_{n=1}^{N} \exp\frac{-SNR_{i,n}}{\beta_m}\right), \tag{3}$$

where α_m and β_m MCS-specific calibration factors [37], and N is the number of PRBs allocated to UE i. The SNR-to-CQI mapping follows the practical estimated channel in [41], and the CQI-to-MCS mapping follows [36]. The selected MCS is then adapted via link adaptation mechanism, based on the application's target PER_i. While practical implementations rely on block error rate (BLER) curves, our simulation uses theoretical bit error rate (BER) models for each MCS (up to 64QAM), assuming a Rayleigh fading channel [42]. To ensure reliable data transmission, especially for critical applications, the system allows up to three retransmission attempts per packet. The simulations support multiple parallel HARQ processes, enabling the transmission of new packets while the previous packets await the HARQ feedback (i.e., four slots). Meanwhile, HOL delay is computed in the higher layers, independent of HARQ, ensuring that the scheduling priorities are based on queue waiting time rather than ongoing retransmissions. Packets that still fail after these attempts are considered dropped due to poor channel conditions. The instantaneous throughput for UE i at time t is defined as

$$TH_i^t = TBS_i^t \cdot nr_{slots} \cdot nr_{frames}, \tag{4}$$

where nr_{slots} is the number of slots per radio frame, and nr_{frames} is the number of radio frames per second.

Our main challenge is the efficient allocation of limited radio resources to multiple users with heterogeneous QoS requirements, while fulfilling their requirements and serving within time constraints. To tackle this, we formulate an optimization problem that maximizes the overall system utility, defined based on each UE's priority and packet urgency:

$$\max_{t,\delta_{i,r}^t,z_{i,k},b_{i,k}} \quad \sum_{i\in\mathcal{I}}\sum_{k\in\mathcal{K}_i}\left(z_{i,k}\cdot\frac{P_i}{\max(\epsilon, PDB_i - D_{i,w}^k)}\right) \tag{5}$$

$$\text{s.t.} \sum_{k\in\mathcal{K}_i} z_{i,k}\cdot pkt_i^{k,bits} \le TBS_i^t, \quad \forall i \tag{5a}$$

$$\sum_{i\in I}\delta_{i,r}^t \le 1, \forall r \tag{5b}$$

$$\sum_{i\in I}\sum_{r\in R}\delta_{i,r}^t \le R \tag{5c}$$

$$z_{i,k} \ge b_{i,k}, \quad \forall i,k \tag{5d}$$

$$b_{i,k} \ge z_{i,k+1}, \quad \forall i,\ k+1 \in \mathcal{K}_i \tag{5e}$$

$$\delta^t_{i,j} \in \{0,1\}, \quad \forall i, \quad \forall r, \tag{5f}$$

$$z_{i,k} \in [0,1], \quad \forall i,k \tag{5g}$$

$$b_{i,k} \in \{0,1\}, \quad \forall i,k \tag{5h}$$

In this formulation, $z_{i,k}$ denotes the fraction of packet k of UE i scheduled for transmission and ϵ is a small positive constant to ensure numerical stability. Constraint (5a) ensures that the total amount of scheduled data remains within the TBS allocated per TTI to each user. The orthogonality of RBG allocation is defined in constraint (5b), while constraint (5c) limits the total number of allocated RBGs to the threshold R. Constraints (5d) and (5e) ensure a FIFO-like scheduling policy, enforcing sequential transmission of packets. The objective aims at the transmission of a large number of packets, prioritizing urgent and high-priority packets, particularly those near their deadlines. Consequently, the allocation decisions in constraint (5a) inherently favor users with better channel conditions, as these users yield higher utility in the objective.

This optimization problem is a mixed-integer programming (MIP) problem due to binary variables $\delta^t_{i,r}$ and $b_{i,k}$, rendering the problem NP-hard and computationally intractable as I scales. To address this, we initially employ an off-the-shelf solver, Gurobi [43], which serves as a theoretical upper bound on achievable performance and is referred to as the Optimal method throughout the paper. In addition, we propose a heuristic algorithm, named QUEST (described in Section V), which reduces the computational complexity of the optimization problem and enables practical implementation in real deployments.

A. User Satisfaction

To quantify the fulfillment of UEs' QoS requirements, we define a user satisfaction metric $\vec{U(i)} = [U^1(i), U^2(i), \cdots, U^T(i)]$ where each element $U^t(i) \in [0,1]$, corresponds to the satisfaction level at TTI. We propose this metric because the existing metrics, such as the interactivity score [44], do not account for throughput and the influence of 5QI parameters. The average $\bar{U(i)}$ captures the satisfaction over the observation window T. A value of $\bar{U(i)} = 0$ indicates a complete failure to meet the QoS demands (i.e., no packet is successfully transmitted). Meanwhile, $\bar{U(i)} = 1$ reflects perfect satisfaction, where all buffered data are transmitted within the PDB_i with minimum delay and without packet drops. Notably, by reducing the packet transmission time and drop rate, *QUEST* shortens the device wake-up time, thereby indirectly increasing energy-efficiency (EE) at the device side. The satisfaction metric comprises of three components: throughput $U_{th}(i)$, delay $U_d(i)$, and packet success ratio $U_{suc}(i)$, combined as

$$\bar{U}(i) = U_{\text{suc}}(i) \cdot \frac{1}{T} \sum_{t=1}^{T} U^t_{th}(i) \cdot U^t_d(i). \tag{6}$$

The throughput and delay components are evaluated at each scheduling slot t, while the packet success ratio is computed at the end of the measurement period or simulation run. The throughput component measures the fraction of data transmitted at time t relative to the buffer state, as

$$U^t_{th}(i) = \frac{\min(TBS^t_i, BSR^t_i)}{BSR^t_i}, \quad \forall t \tag{7}$$

If the allocated TBS^t_i exceeds the buffer size BSR^t_i, the throughput component is limited by the buffer size. The delay component captures the portion by which the packet transmission delay remains below the PDB_i.

$$U^t_d(i) = \frac{\sum_{k \in K_i} z_{i,k} \cdot \left(1 - \frac{D^k_{i,w}}{PDB_i}\right)}{\sum_{k \in K_i} z_{i,k}}, \quad \forall t \tag{8}$$

The packet success ratio component reflects the portion of packets delivered without violating latency or reliability constraints

$$U_{suc}(i) = \frac{1}{|K_i|} \cdot \sum_{k \in K_i} z_{i,k} \tag{9}$$

The user satisfaction metric is applied to the data collected from the experimental hardware setup and simulation. Unlike classical fairness metrics, such as Jain's index [28], the user satisfaction metric captures fairness in a heterogeneous service environment. It ensures that users are treated fairly relative to their QoS requirements, rather than focusing on a single performance metric, such as throughput. Starvation of a user is represented by a satisfaction value being zero.

V. QUEST Design and Implementation

In this section, we introduce *QUEST*, a heuristic scheduling framework that leverages both environment context information (i.e., time-frequency selectivity of wireless channel), and user context information (i.e., application QoS class and HOL delay) to efficiently allocate scarce radio resources, in an indoor factory environment supporting heterogeneous 5QI services. To ensure seamless integration into existing 5G deployments, *QUEST* is designed as heuristic-based scheduling and incorporates our enhancements to standardized signaling mechanisms. As a result, it avoids the computational overhead associated with complex optimization techniques, making it particularly suitable for the strict latency and real-time processing requirements of 5G networks. *QUEST* is fundamentally driven by the 5G QoS model (see III-A), resulting in a scheduling metric not only channel-aware but also application and delay-aware. Enabling delay-awareness in UL is a non-trivial challenge, which *QUEST* addresses by enhancing the existing BSR signaling, with HOL delay information.

A. HOL Delay Reporting

The waiting time of queued packets provides the scheduler with insights into UEs whose packets have not been recently scheduled. The information enables the scheduler to prioritize packets experiencing high waiting time, especially those approaching their application-specific delay constraint. While reporting per-packet delay $D^t_{i,k}$ provides fine-grained information, it introduces considerable signaling overhead. Therefore, the HOL delay, representing the maximum queue

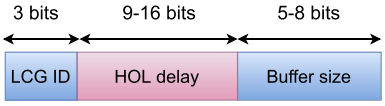


Fig. 4. Short BSR MAC CE, with integrated HOL delay field for a LCG. The same applies to the long BSR MAC CE; however, it expands for multiple LCGs.

delay, is used as an efficient alternative. However, unlike DL approaches [10], [28], [45], integrating HOL delay reporting in the UL poses a significant challenge, due to the absence of a dedicated signaling mechanism.

To overcome this limitation, *QUEST* introduces a custom enhancement of existing BSR signaling by adding an additional field dedicated to HOL delay, as illustrated in Fig. 4. This field is defined as a 16 bit extension with 10 μs granularity, covering delays up to 300 ms, sufficient for all 5QI PDB requirements. Although high precision is essential for delay-critical applications, there is a trade-off between precision and signaling overhead. For instance, relaxing the precision to 1 ms reduces the field size to 9 bit. Given that the BSR messages carry information for multiple LCGs, the incremental addition of the HOL delay field results in a marginal increase in the overall control message size. To mitigate its overhead in dense scenarios with frequent reporting, adaptive reduction of the precision can be employed, balancing the need for accuracy against the signaling efficiency.

We note that the HOL delay reporting mechanism reuses the BSR MAC control element (CE) and the respective logical channel identifier (LCID) index [39], while extending its payload. This design requires coordinated support at both the UE and gNB to ensure correct parsing of the extended BSR format. Consequently, the proposed mechanism is not intended to interoperate transparently with unmodified commercial off-the-shelf devices, but is suitable for experimental platforms, private networks, or vendor-controlled deployments. We have implemented and validated this extension within our experimental hardware framework, confirming that it preserves correct MAC-layer operation when supported by both endpoints.

B. QUEST Heuristic Metric

QUEST heuristic approach proposes a scheduling metric that incorporates the main factors influencing the scheduling decisions, as defined in [28]: (1) enhances UEs' instantaneous data rate and reduces the reliability violations, by exploiting the frequency selectivity of the wireless channel during RBG allocation; (2) reduces delay violations by prioritizing UEs with HOL delays approaching their application PDB; (3) prioritizes packets coming from higher priority level applications despite having the same PDB. The proposed *QUEST* scheduling metric for UE i at RBG r, part of subband s, during time t is defined as

$$\psi^t_{i,r} = \frac{P_{max}}{P_i} \cdot \frac{TBS^t_{i,r}}{max(\epsilon, PDB_i - D^t_{i,HOL})}, \forall i, \forall r, \forall t, \qquad (10)$$

where the P_i and PDB_i are the priority level and the packet delay budget of the application, respectively. Since the 5G QoS model [13] assigns lower integer values to higher priorities P_i, the metric is scaled by the maximum priority value $P_{max} = 90$, to balance across priority classes. The instantaneous $TBS^t_{i,r}$ reflects the channel quality of the respective subband. Delay violations, where the HOL delay $D^t_{i,HOL}$ exceeds the packet delay budget PDB_i, are mitigated by employing the $max()$ function, in which a small constant ϵ is used to ensure numerical stability. This approach ensures that the delay violations are appropriately penalized to avoid further packet drops.

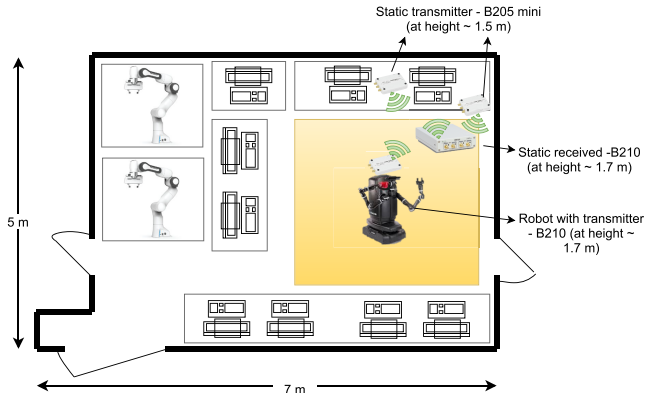


Fig. 5. Measurement scenario in an indoor environment, with three static transmitters (UEs) and a static receiver (gNB).

The *QUEST* metric is computed independently for each UE i and each RBG r at every scheduling interval t. Therefore, for I connected UEs and R RBGs, the per-slot complexity is $O(I \cdot R)$, linear with the number of UEs and RBGs. The *Optimal* method has a worst-case complexity of $O(2^{I \cdot (R+K)})$, exponential with the number of UEs I, packets per user K, and radio resources R. The exponential growth makes the real-time computation infeasible even for a moderate number of UEs and radio resources.

VI. Measurement Setup and Implementation

The performance of *QUEST* is evaluated not only in simulations but also in a measurement setup of a 5G network. In this section, we initially describe the measurement environment as well as the utilized hardware, followed by the data acquisition and description of implemented UL scheduling metrics.

A. Measurement Setup and Hardware

The measurements were conducted in an indoor environment, featuring multiple robotic arms, mobile robots, and PCs, representing a typical indoor factory scenario, as illustrated in Fig. 5. The experimental setup is based on the OAI software [46], implementing a 5G network compliant with 3GPP standard. The scenario includes one receiver (gNB) and three transmitters (UEs), all using SDRs from Ettus universal software radio peripheral (USRP) with omnidirectional antennas. The static transmitters are deployed on B205-mini USRPs connected to computers with an Intel(R) Core(TM) i7-6700 CPU @ 3.40GHz, running Linux 6.8.0-45-generic kernel. Meanwhile, the robot transmitter [47], [48] is deployed on B210 USRP connected to a computer with an Intel(R) Core(TM) i7-8559U CPU @ 2.70GHz processor with the Linux 5.4.0-48-generic kernel. The gNB receiver is deployed on a B210 Ettus USRP connected to a computer with an

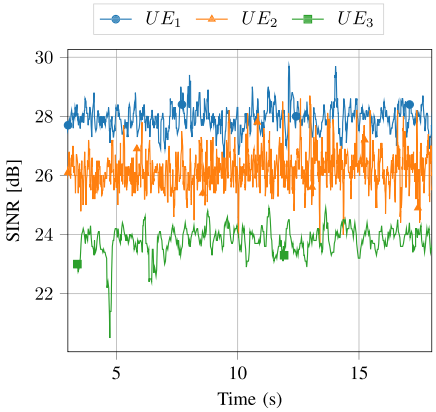


Fig. 6. Illustration of the measured channel during one measurement set for three UEs in the indoor factory scenario.

Intel(R) Core(TM) i7-4790 CPU @ 3.60GHz processor with the Linux 5.15.0-122-generic kernel.

All devices use the open-source USRP hardware driver (UHD) [49] for continuous data transmission and reception. Transmit and receive gains are identical across devices. Communication occurs in 5G NR band n78 at carrier frequency 3.6192 GHz, over 40 MHz bandwidth, 30 kHz subcarrier-spacing and a time division duplex (TDD) configuration 3:1:1 (DDDSU). The setup uses wideband CQI for computing the channel quality.

B. Data Acquisition

The proposed *QUEST* framework extends the transmitter's side implementation to include HOL delay information within the BSR reports. Correspondingly, we extend the receiver-side implementation to: (1) extract and process the reported HOL delay information by UEs; (2) retrieve the relevant QoS flow parameters through configured DRBs in the MAC, and then parse them within the UL scheduler; (3) capture the UL transmission delay of received packets in UL scheduler; (4) implement multiple scheduling algorithms, including MT, PB, EDF, and M-LWDF; (5) evaluate *QUEST* framework.

Each scheduling approach is evaluated over five independent measurement sets, each lasting 30 s. Between sets, the transmitters' positions change relative to the gNB to capture diverse channel conditions. Within each run, all devices remain static to ensure stable channel conditions for fair comparison across schedulers. Channel conditions vary over time, as shown in Fig. 6, due to reflections from metallic objects, irregular object shapes, and blockages. Due to the TDD configuration in OAI, UEs transmit during the shared slots (7 and 17) and dedicated UL slots (8, 9, 18, and 19) in each frame.

To emulate realistic traffic patterns, especially those of delay-critical applications, we use the isochronous round-trip tester (IRTT) tool [50]. The IRTT clients run at the transmitters while the IRTT server operates at the CN. This setup generates periodic UDP packets, representative of a deterministic flow used in applications such as discrete automation of V2X communication. The original IRTT tool records UL, DL, and round-trip delays for successfully transmitted packets at the end of each run. To enable real-time access to the UL delay data in the scheduler, we modified the IRTT tool, to include additional timestamps, allowing the IRTT server to capture precise one-way UL delay and forward it to the MAC scheduler.[1] To synchronize the IRTT server and clients' applications, the precision time protocol (PTP) was used with the hardware timestamp. It was operated out-of-band to avoid any interference between measurement traffic and synchronization, thereby preserving its accuracy. The PTP source runs at CN, while the PTP sink operate at UEs. The PTP synchronization achieves sub-microsecond precision [51], ensuring accurate UL measurements.[2]

C. Implementation of Scheduling Algorithms

We evaluate the resource scheduling efficiency of *QUEST* against representative state-of-the-art algorithms. In addition to the existing PF implementation in the OAI setup, other baseline schedulers have been integrated at the receiver side (gNB MAC layer). Specifically, we include MT alongside PF as representative of channel-aware scheduling algorithms and PB, EDF, and M-LWDF as representative of QoS-aware scheduling algorithms. At time t, a RBG r is assigned to the UE i achieving the highest scheduling metric $\psi_{i,r}^t$ among connected users, defined as

$$i^* = \arg \max_{i \in \{1,2,\ldots,I\}} \psi_{i,r}^t \tag{11}$$

Maximum throughput (MT): To maximize the total instantaneous data rate of the system, MT allocates the available RBG r to the UE i experiencing the best channel conditions. Since better channel conditions result in higher TBS, the scheduling metric for UE i at RBG r during time t is

$$\psi_{i,r}^t = TBS_{i,r}^t. \tag{12}$$

Proportional fair (PF): Introduces throughput fairness in channel-aware scheduling [55]. The objective is to maximize the total achieved data rate while maintaining throughput fairness across users. The scheduling metric is weighted by the UE's historical allocation until current time T.

$$\psi_{i,r}^t = \frac{TBS_{i,r}^t}{\sum_{t \in T} TH_i^t}. \tag{13}$$

The relative weight of $TBS_{i,r}^t$ and historical allocation TH_i can be adjusted [30]. However, default values are used in the baseline implementation.

Priority-based (PB): Assigns RBGs based solely on the packet priority level specified in the 5G QoS model [24]. Unlike MT and PF, it is agnostic to channel conditions and relies exclusively on P_i for scheduling decisions.

$$\psi_{i,r}^t = \frac{P_{max}}{P_i}. \tag{14}$$

Earliest delay first (EDF): Prioritizes packets based to their proximity to *PDB*. While effective at prioritizing real-time traffic, the lack of channel awareness degrades the performance

[1]The delay between the CN and gNB is negligible, orders of sub- ms, compared to the UL delay between UE and gNB.

[2]Negative UL delays are recorded if not utilizing PTP synchronization.

under dynamic conditions. Given a delay budget PDB_i and the HOL delay, the metric is defined as

$$\psi_{i,r}^t = \frac{1}{max(\epsilon, PDB_i - D_{i,HOL}^t)}. \quad (15)$$

Modified-largest weighted delay first (M-LWDF): A widely used QoS-aware scheduling algorithm, aware of channel conditions, delay requirements [10], [18]. For UE i at time t, given a delay threshold PDB_i, delay violation probability γ_i, and current HOL delay $D_{i,HOL}^t$, the scheduling metric is

$$\psi_{i,r}^t = -\log(\gamma_i) \cdot \frac{D_{i,HOL}^t}{PDB_i} \cdot \frac{TBS_{i,r}^t}{\sum_{t \in T} TH_i}. \quad (16)$$

By integrating these well-established and widely accepted scheduling algorithms as our baselines, we ensure a comprehensive comparison across main scheduling categories, channel-aware, priority, and QoS-aware, and demonstrate the improvements of *QUEST* in the UL resource allocation.

VII. Performance Evaluation

In this section, we evaluate the performance of *QUEST* using both testbed measurements and large-scale simulations. Our evaluations focus on packet delay, packet error rate, and user satisfaction. We benchmark *QUEST* against diverse state-of-the-art schedulers, including channel-aware schedulers such as PF and MT, as well as QoS-oriented schedulers such as PB, EDF, and M-LWDF, described in Section VI-C. Given that the evaluation is conducted in UL, we apply our proposed HOL delay reporting mechanism to all schedulers. This adaptation enables their operation in the UL without modifying their inherent scheduling logic. These algorithms are evaluated on applications commonly encountered in an indoor factory environment, with traffic patterns and 5G QoS characteristics summarized in Table II for both measurements and simulation.

Some applications appear in both setups, while others are selectively varied to show the effectiveness of *QUEST* across use cases. The selected applications include diverse requirements, including relaxed PDB and low priority for augmented reality (AR), stringent PDB and moderate priority for control-to-control communication, and relaxed PDB with high priority for mobile robots. The device density reflects the portion of the total devices associated with each application. Note that each device is assigned a single application per run.

For measurements, we use the IRTT traffic generator, configured with fixed packet sizes and IAT selected from the ranges in Table II. The results are presented using aggregated cumulative distribution functions (CDFs) and boxplots, based on five datasets, each collected over 30 s measurement time, representing different scenarios. These scenarios reflect diverse channel conditions and spatial positions of UEs relative to gNB, ensuring that the observed performance extends beyond a specific scenario and is consistent across different network environments. Within each dataset, the scenario remains fixed while the resource scheduling methods are evaluated.

To further demonstrate the advantages of *QUEST* and ensure its generalization in larger scenarios, we conduct seeded simulations of an indoor factory scenario using the NextGSIM simulator [27], extended within a Python 3.8-based framework.

TABLE I
List of Acronyms and Symbols

Abbreviation	Definition
5QI	5G QoS identifier
AMC	adaptive modulation and coding
BSR	buffer status report
CN	core network
CQI	channel quality indicator
DB	delay based
DRBs	data radio bearers
EDF	earliest delay first
HOL	head-of-line
IAT	inter-arrival time
MDBV	maximum data burst volume
M-LWDF	modified-largest weighted delay first
MT	maximum throughput
PB	priority based
PDB	packet delay budget
PER	packet error rate
PF	proportional fair
QFI	QoS flow identifier
QoS	quality of service
RBG	resource block group
RRM	radio resource management
SDAP	service data adaption protocol
SDR	software defined network
TBS	transport block size
TDD	time division duplex
UPF	user plane function
USRP	universal software radio peripheral
i	UE index
k	packet index
r	RBG index
t	TTI index
s	subband index
K_i	set of packets of UE i
$\psi_{i,r}^t$	scheduling metric
$pkt_i^{k,bits}$	packet size
$pkt_i^{k,t}$	packet arrival time
D_i^k	packet service time
$D_{i,w}^k$	packet waiting time
$D_{i,HOL}^t$	user head of line delay
P_i	packet priority level
P_{max}	maximum priority value
PDB_i	packet delay budget
PER_i	packet error rate target
BSR_i^t	buffer status report (buffer size)
$\bar{U}(i)$	average user satisfaction
TBS_i^t	total transport block size at time t
$TBS_{i,r}^t$	transport block size on RBG r
TH_i	user's instantaneous throughput
$CQI_{i,s}^t$	channel quality indicator
$MCS_{i,s}^t$	modulation and coding scheme
$Q_{m,i,s}^t$	modulation order
$CR_{i,s}^t$	coding rate
N_{RE}	number of resource elements
v	number of MIMO layers
$z_{i,k}$	fraction of packet k of UE i scheduled
$b_{i,k}$	binary variable enforcing packet order (FIFO)
$\delta_{i,r}^t$	RBG allocation indicator
nr_{slots}	number of slots per frame
nr_{frames}	number of frames per second
ϵ	small positive constant
T_{GBR}	GBR observation interval
GBR_i	target bit rate

We incorporate our system model (Section IV), the previously described schedulers (Section VI-C), and the *QUEST* framework into the simulator. In this environment, we study the scalability and the performance of *QUEST* (in Section VII-B) under highly dynamic conditions, where mobile users follow a random waypoint mobility. UEs are initially distributed

TABLE II
INDOOR FACTORY APPLICATIONS CHARACTERISTICS AND REQUIREMENTS [13], [52], [53], [54]

Use case		Application	5QI	Default Priority	Traffic Type	Traffic Characteristics Packet Size (bytes) and Packet IAT (ms)	PDB (ms)	PER	Device Density
Simulation		Discrete automation	82	19	Periodic	Uniform($min = 200, max = 255$) Uniform($min = 8, max = 10$)	10	10^{-4}	0.24
		V2X messages	83	22	Periodic	Uniform($min = 1000, max = 1345$) Uniform($min = 9, max = 10$)	10	10^{-4}	0.16
	Measurement	Augmented reality	2	40	Bursty	Truncated Pareto($\alpha = 1.2, min = 800, max = 1200$) Truncated Pareto($\alpha = 1.1, min = 2.5, max = 12.5$)	150	10^{-3}	0.4
		Mobile robots	69	5	Aperiodic	Uniform($min = 40, max = 250$) Uniform($min = 1, max = 30$)	60	10^{-6}	0.2
		Control-to-control	84	24	Periodic	Uniform($min = 1000, max = 1345$) Uniform($min = 12, max = 15$)	30	10^{-5}	0.4

TABLE III
OVERVIEW OF SIMULATION PARAMETERS

Parameters	Value
Simulation time	30 s
System bandwidth (W)	40 MHz (106 PRBs)
Subcarrier spacing (σ)	30 kHz
Max HARQ retransmission	3
Carrier frequency	3.6192 GHz
Number of gNBs	1
Channel model	Indoor factory SL channel [58]
Channel estimation granuarity	wideband, subband, PRB
Resource allocation granularity	RBG[8], PRB[1]
Number of users	50

randomly, and move based on individual velocity vectors (constant during each simulation run but varying across users). Considering the cell area of the indoor factory scenario, and the expected device density of up to $10^6/km^2$ mMTC devices [56], approximately 80 mMTC devices can connect to a cell of this size. However, these devices generate sporadic, small-payload traffic and impose a lower scheduling load than latency-critical services. Due to the heterogeneity of the considered applications and their diverse traffic patterns, 50 UEs represent a realistic load for a small cell in the studied indoor factory scenario.

The simulation setup mirrors the measurement environment, with a single gNB serving multiple users through the 3GPP-compliant indoor factory SL model [57]. Table III outlines the key simulation parameters. While wideband channel estimation is used for consistency with measurements, we also simulate subband and PRB-level channel estimation to better capture the frequency selectivity of the wireless channel. Although the PRB-level channel estimation and resource allocation granularity are ideal and computationally intensive, we use them to benchmark scheduler performance against the *Optimal* method in Section VII-C.

In the selected applications, we highlight the distinction between discrete automation and V2X traffic. Despite their identical PDB, they have different scheduling priorities, providing an insightful case for performance differentiation. Simulation results are presented as boxplots grouped by 5QI classes, where values represent the average performance of the UEs associated with each application type.

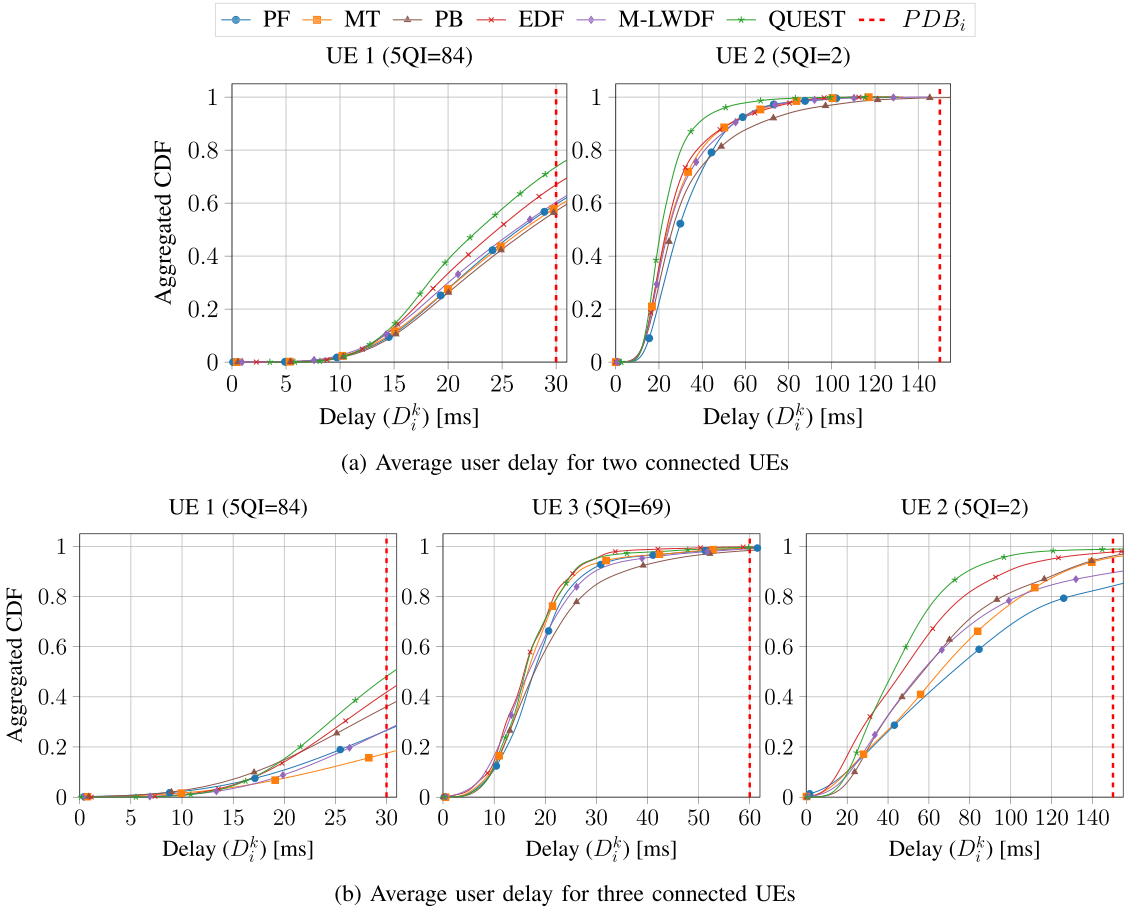
A. Performance of Users in Measurements

Fig. 7a shows the CDFs of the packet serving delay D_i^k for two users running AR and control-to-control messaging applications. Measurements were conducted in a congested network with insufficient radio resources to fully serve all connected UEs. The results demonstrate that *QUEST* consistently achieves the lowest packet delays while transmitting the most packets within UEs' PDB. This comes from *QUEST*'s ability to jointly consider instantaneous TBS, UL HOL delay, and its proximity to PDB, thus balancing the performance of UEs.

For UE 1, with stringent delay requirements, *QUEST* reduces the average delay by 17.5% (4.86 ms) for 50% of the transmitted packets compared to the worst-performing state-of-the-art (SoA) scheduler, PB. When compared to the best-performing SoA scheduler, EDF, the delay is reduced by 6.2% (1.52 ms). Similarly, for UE 2, *QUEST* decreases the delay for 90% of transmitted packets by 42.4% (28.5 ms) relative to PB, and by 26.8% (14.2 ms) compared to MT, which is the best-performing SoA in this case. Network congestion limits UE 1 to transmitting only a subset of its packets, due to limited radio resources, stringent PDB, and large packet sizes.

In this scenario, *QUEST* outperforms the other schedulers by serving 74.7% of the packets within the PDB, followed by EDF (66.2%),M-LWDF (61%), PF (59.3%), MT (57.7%), and PB (56.4%). This advantage is reflected in the user satisfaction metric (Fig. 8a), where *QUEST* shows superiority for both connected UEs. By scheduling efficiently, *QUEST* reduces the active transmission time and overall energy consumption of UEs, making it suitable for resource-constrained IoT devices.

To assess the impact of application priority, we introduce a mobile robot application $5QI = 69$, which, despite its relatively relaxed PDB of 60 ms, has the highest priority. Due to its longer inter-arrival time (IAT) and small packet size 250 byte, all schedulers are able to fully serve the packets generated by UE 3, as shown in Fig. 7b. However, the lack of channel awareness in PB results in degraded performance despite assigning resources based on priority. The presence of high-priority UE significantly affects schedulers that target throughput fairness, such as PF and M-LWDF. In this case, *QUEST* continues to maintain the best performance, while inevitably reducing UE 1 and UE 2 satisfaction due to increased competition for resources, as depicted in Fig. 8b. Specifically, UE 1 experiences a 31% increase in delay and a 35% reduction in successfully transmitted packets compared to the scenario with only two connected UEs. Nevertheless, *QUEST* still achieves on average a 4.5% increase in user satisfaction for UE 1 and a 4.9% increase for UE 2 compared to EDF, confirming its superiority even with a higher number of UE and more diverse applications.



(a) Average user delay for two connected UEs

(b) Average user delay for three connected UEs

Fig. 7. Average CDF of UEs' delays measured for (a) two users and (b) three users running different applications.

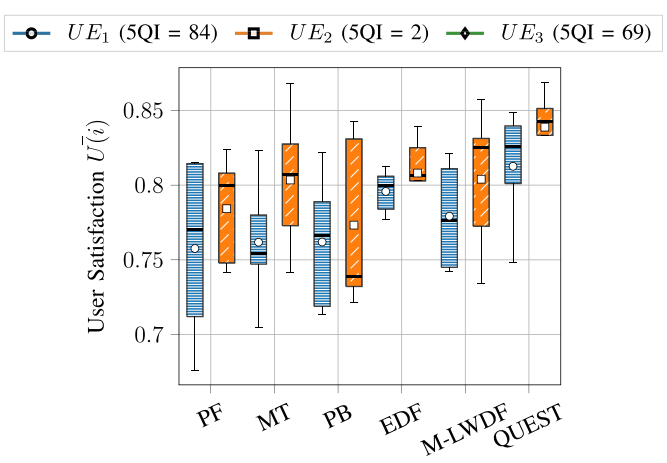
B. Scalability of QUEST

We evaluate the scalability of *QUEST* through simulations that incorporate realistic channel estimation and resource allocation granularity. Fig. 9, compares user satisfaction achieved with wideband and subband CQI. In both cases, resource allocation follows the standardized RBG granularity, where each RBG comprises eight consecutive PRBs. Given the large number of connected UEs, user satisfaction is aggregated per 5QI class and averaged across the UEs associated with each application type. Fig. 9a shows a reduced performance across all scheduling algorithms and 5QI services, due to the flat channel estimation in the wideband scenario. This effect is pronounced for 5QI 83, characterized by frequent, large packets. It experiences frequent packet drops due to PDB violation and poor channel conditions, particularly when resources are allocated to applications with higher scheduling metrics. Among baseline algorithms, PB and EDF outperform traditional throughput-based schedulers. While *QUEST* notably increases the user satisfaction for the highest-priority 5QI 69, it slightly reduces that of 5QI 82 and 83. This highlights the tradeoff of resource allocation under congestion, where prioritizing certain applications comes at the cost of performance for other high-demand applications.

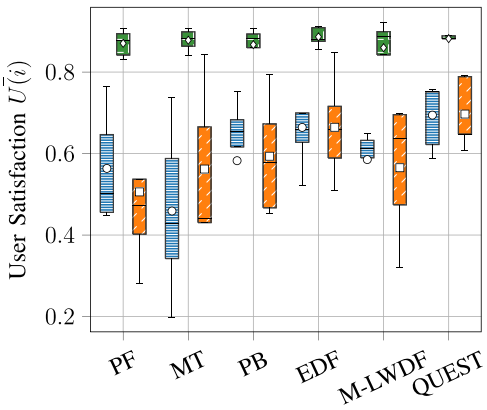
In contrast, subband CQI enables schedulers to better exploit frequency selectivity, thereby improving overall performance, as illustrated in Fig. 9b. The advantages of PB and EDF seen in the wideband case diminish here. While M-LWDF records the lowest packet drops among the baselines, its average packet delays reduce the user satisfaction (i.e., longer times to transmit packets within the 5QI requirements). To illustrate the scalability, we extend the subband CQI scenario to also cover 100 UEs in Fig. 9c. *QUEST* maintains its performance under increased load, though high contention reduces average user satisfaction across all 5QI classes. The low user satisfaction for 5QI 2 is related to serving the packets closer to their PDB. In this scenario, we emphasize that our proposed framework achieves the best performance, particularly for UEs with stringent delay and higher-priority applications with 5QI [69, 82, 83]. These results support the effectiveness of the framework as an EE resource scheduling framework.

C. Evaluation Against Optimal Method

Fig. 10 compares the performance metrics of baseline schedulers, *QUEST*, and the Optimal method, which is described in Section IV. The results correspond to an ideal scenario with PRB level channel quality information and per-PRB level resource allocation granularity. Nevertheless, we emphasize that even in such fine-granularity, the UL packet transmissions are carried out using a single MCS for UE. As expected, the Optimal method achieves the highest user satisfaction, efficiently transmitting packets with minimal delay within PDB for all 5QI applications, as illustrated in Fig. 10a. It achieves this satisfaction without dropping any packets, as shown in Fig. 10b, which presents the PER on a logarithmic scale. This



(a) User satisfaction for two connected UEs



(b) User satisfaction for three connected UEs

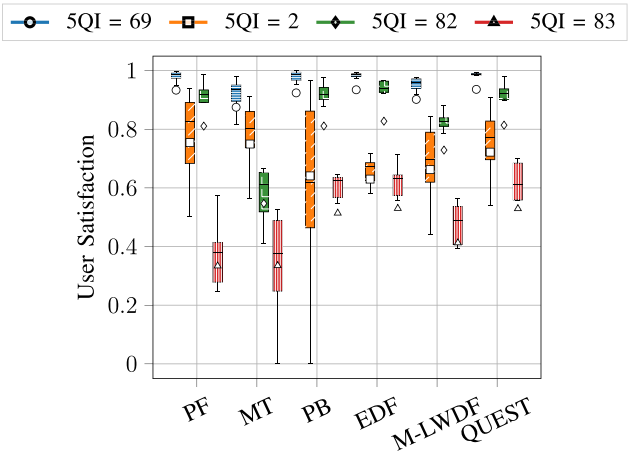
Fig. 8. Measured user satisfaction in two district scenarios.

Optimal method is followed by *QUEST*, which is the closest approximation among the evaluated heuristics. It successfully meets the target PERs, with packet drops occurring only for 5QI 2 (see Fig. 10b). Unlike the wideband and subband scenarios, the MT scheduler also performs well in this scenario, achieving a good performance in user satisfaction and PER. This improvement is attributed to the increased exploitation of channel variations, enabled by PRB-CQI, which reduces the congestion encountered in the simulation scenario.

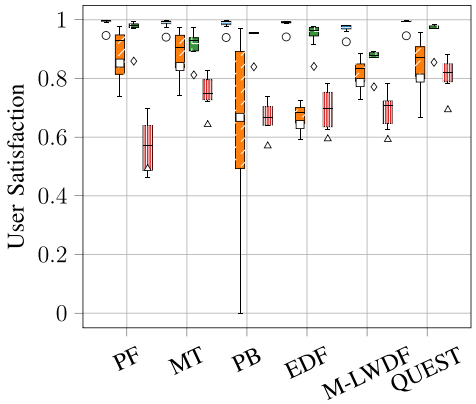
We also compare runtimes under a scenario with 50 connected UEs. The Optimal method uses the Gurobi solver via its Python API, with the core solver written in C. Despite this, the average execution time of the Optimal method is 244 ms. In contrast, *QUEST* achieves a substantial runtime reduction of approximately 84%, with an average execution time of 38 ms, in the Python-based simulation. The C++ implementation of *QUEST* in our testbed shows an average execution time of approximately 0.035 ms for 3 UEs. While the testbed scenario differs in scale from simulations, the results demonstrate the suitability of *QUEST* for real-time deployment.

VIII. Discussion

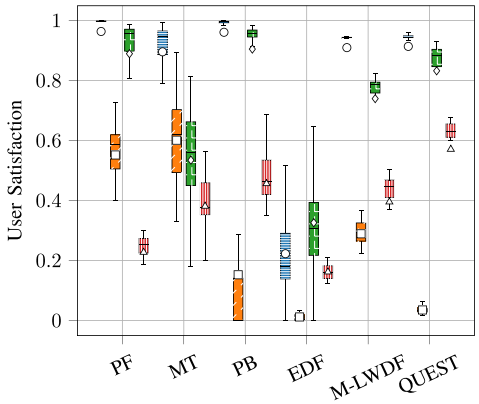
In this section, we discuss potential deployment aspects and their potential impact on *QUEST*.



(a) Wideband CQI, 50 UEs



(b) Subband CQI, 50 UEs

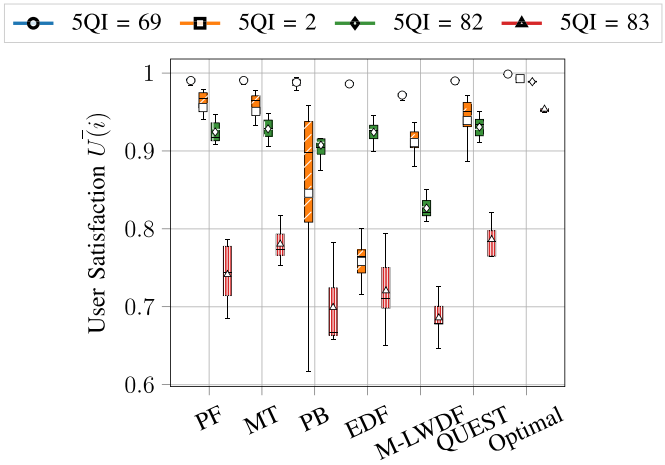


(c) Subband CQI, 100 UEs

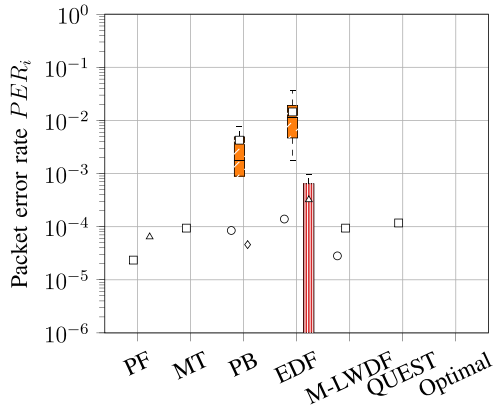
Fig. 9. Comparison of user satisfaction in simulated scenarios.

A. Inter-Cell Interference

In dense deployments or at cell edge, UL transmissions from neighboring cells can degrade the channel quality experienced by the UEs. Within the context of *QUEST*, inter-cell interference directly affects the instantaneous TBS term in the scheduling metric (see Section V-B), thereby reducing the prioritization of UEs experiencing severe interference. Consequently, UL inter-cell interference indirectly increases the perceived packet delays for affected users. Nevertheless, since the *QUEST* metric incorporates both UEs' HOL delay and application-specific QoS requirements, it mitigates user



(a) Average user satisfaction



(b) Average packet error rate

Fig. 10. Performance metrics in simulation scenario, benchmarked against Optimal method.

starvation by increasing scheduling priority as the HOL delay approaches the corresponding PDB threshold.

B. Multi-Cell Scheduling

UL multi-cell scheduling relies on inter-cell interference coordination mechanisms, such as power control signaling, coordinated resource partitioning, to limit simultaneous transmission on overlapping RBGs, particularly for cell-edge UEs. This coordination reduces the number of RBGs available at a given transmission interval, thereby constraining the scheduling flexibility. Nevertheless, *QUEST* remains applicable in such scenarios, as it adapts to the reduced resource pool by prioritizing the UEs with better channel quality, stringent packet delays, and high HOL delays.

C. Multiple QoS Flows in One DRB

Although the current evaluation considers a single application per UE, *QUEST* is not inherently restricted to this assumption. When multiple QoS flows are multiplexed onto the same data radio bearer (DRB), a single logical buffer at the resource link control (RLC) layer aggregates traffic across flows, thereby obscuring per-flow queue dynamics. As a result, the flow-level HOL delay is not directly observed at

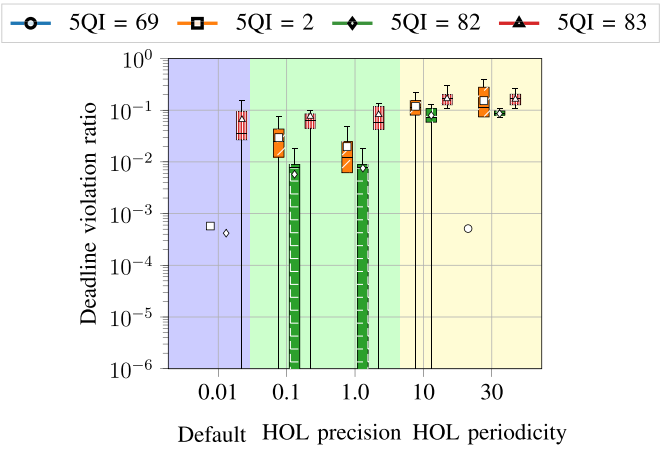


Fig. 11. Sensitivity analysis with respect to HOL delay quantization and HOL reporting periodicity for 100 UEs.

the RLC buffer. Instead, the per-flow HOL delay is calculated at higher protocol layers and reported via the BSR as the maximum delay among all active flows mapped to the DRB. This ensures that the most delay-sensitive flow drives scheduling decisions, while lower-priority flows are opportunistically scheduled. Although this aggregated approach reduces per-flow granularity, it maintains delay-aware prioritization and QoS differentiation.

D. HOL Delay Sensitivity Analysis

We conduct a sensitivity analysis with respect to the (i) HOL delay quantization and (ii) HOL delay reporting periodicity. For the HOL quantization we evaluate three representative precisions 10 μs (default), 100 μs and 1 ms while keeping the default reporting periodicity of 1 ms. For the reporting periodicity we evaluate three values, 1 ms (default), 10 ms and 30 ms, while keeping the default 10 μs precision. To clearly represent the effects of HOL quantization, we congest the network with 100 connected users. The sensitivity analysis in Fig. 11 examines the fraction of dropped packets due to PDB violations. The results show that finer HOL quantization and more frequent HOL delay reporting reduce the number of deadline violations. In particular, the increase in HOL reporting periodicity is accompanied by a higher deadline violation ratio than the increase in the HOL delay quantization. This occurs because the scheduler receives less accurate, less timely delay information, which masks the urgency of delay-critical packets. Meanwhile, the increase in the HOL delay quantization is visible for users with stringent PDB requirements (5QI 82 and 83), for which the deadline violation ratio increases under coarser quantization.

E. Resource Type Awareness

The proposed *QUEST* scheduling metric operates at the per-TTI level and can be extended to support: (i) GBR, (ii) non-GBR, and (iii) delay-critical GBR (DC-GBR) flows. To account for long-term GBR requirements, the metric is adapted based on the throughput shortfall relative to the target bit rate GBR_i, defined as

$$\Delta TH_i^{GBR}(t) = \max(0, GBR_i - TH_i^{GBR}(t)), \tag{17}$$

where, $TH_i^{GBR}(t)$ denotes the sliding-window average throughput over a GBR observation interval T_{GBR} given by

$$TH_i^{GBR}(t) = \frac{1}{T_{\text{GBR}}} \sum_{\tau=t-T_{\text{GBR}}+1}^{t} TBS_i^{\tau}, \tag{18}$$

where TBS_i^{τ} is the TBS allocated to UE i at time τ.

For **GBR** flows, the metric is defined as

$$\psi_{i,r}^{t,\text{GBR}} = \psi_{i,r}^{t} \left(1 + \alpha \cdot \frac{\Delta TH_i^{GBR}(t)}{GBR_i}\right) \tag{19}$$

with α controlling the emphasis on meeting the GBR. For **non-GBR** flows, the metric stays as $\psi_{i,r}^{t,\text{non-GBR}} = \psi_{i,r}^{t}$.

For **DC-GBR** flows, the HOL delay is included via

$$\psi_{i,r}^{t,\text{DC-GBR}} = \frac{\psi_{i,r}^{t,\text{GBR}}}{\max(\epsilon, PDB_i - D_{i,HOL}^{t})}. \tag{20}$$

The proposed factors boost the scheduling priority of UEs not fulfilling bit rate requirements, while preserving the per-TTI considerations of *QUEST* metric.

IX. Conclusion

In this work, we introduced *QUEST*, a novel QoS-aware UL scheduling framework designed for efficient RRM in 5G and beyond networks supporting heterogeneous 5QI applications. Fundamentally driven by the 5G QoS model, *QUEST* defines a heuristic scheduling metric that jointly leverages environmental context (i.e., channel quality) and user context (i.e., application-specific QoS demands and HOL delay). To enable precise delay-aware scheduling, *QUEST* extends the standard BSR mechanism by incorporating HOL delay as an additional signaling field. We validated the practicality of *QUEST* through its integration in a real-world 5G testbed. Our evaluations, using both measurement data and a 3GPP-compliant system-level simulator in a multi-user indoor factory scenario, demonstrate that *QUEST* significantly improves QoS satisfaction, reducing the packet transmission time and drop rates. These improvements lead to higher device EE. For measurements involving two 5QI services, *QUEST* achieves up to 13% increase in successfully transmitted packets and a 6.2% reduction in delay for 50% of packets for the higher-priority 5QI, compared to the best-performing baseline. Furthermore, benchmarking against the Optimal solution shows that *QUEST* delivers the closest performance. Future work will integrate *QUEST* within network slicing architecture, to further enhance RRM in dynamic, application-driven 5G and beyond networks.

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